

Vigenere Cipher Cryptanalysis

Kasiski Examination, Index of Coincidence and Frequency Analysis

Project Report

Project	Vigenere Cipher Cryptanalysis
Language	C++
Attack Type	Statistical Cryptanalysis
Techniques	Kasiski, IC, Frequency Analysis, Chi-Square
Recovered Key	AMBROISETHOMAS
Key Length	14
Ciphertext Length	395 characters

1. Introduction

The Vigenere Cipher is a classical polyalphabetic substitution cipher. It uses a repeating keyword to apply different Caesar shifts to successive characters. This project demonstrates how statistical cryptanalysis can recover the key and plaintext when the key is unknown.

2. Objectives

The objectives are to clean the ciphertext, find repeated patterns, calculate distances and factors, estimate the key length, calculate Index of Coincidence, split the ciphertext into groups, perform frequency analysis, recover the key, decrypt the message, and verify the result.

3. Kasiski Examination

Repeated sequences of three to five characters are located in the ciphertext. The distances between their occurrences are calculated. Common factors of these distances provide candidate key lengths because repeated distances are often multiples of the repeating key length.

4. Index of Coincidence

The Index of Coincidence measures the likelihood that two randomly selected characters are identical. The ciphertext is divided into groups for possible key lengths and the average IC is calculated. A suitable key length produces groups with statistics closer to natural English text.

5. Frequency Analysis

After estimating the key length, the ciphertext is split into groups. Each group behaves approximately like a Caesar cipher. Letter frequencies are compared with standard English frequencies for all possible shifts.

6. Chi-Square Shift Detection

For every possible shift from 0 to 25, a chi-square score is calculated between observed and expected English letter frequencies. The shift with the smallest score is selected for each group.

7. Key Recovery

The best shift from each group is converted into a letter. Combining the letters gives the recovered Vigenere key. For this ciphertext, the recovered key is AMBROIETHOMAS and its length is 14.

8. Decryption

The recovered key is used with the Vigenere decryption operation $P = (C - K) \bmod 26$ to obtain the plaintext.

9. Verification

The recovered plaintext is encrypted again using the recovered key. The generated ciphertext is compared with the original cleaned ciphertext. An exact match confirms the result.

10. Important Functions

Function	Purpose
clean_ciphertext()	Removes spaces and non-alphabetic characters.
find_repeated_patterns()	Finds repeated patterns of length 3 to 5.
calculate_distances()	Calculates distances between repeated occurrences.
find_factors()	Finds factors of the calculated distances.
kasiski_analysis()	Counts factors to estimate possible key lengths.
calculate_ic()	Calculates the Index of Coincidence.
split_into_groups()	Divides ciphertext according to the key length.
frequency_analysis()	Displays letter frequencies for each group.
find_shift()	Finds the best Caesar shift using chi-square.
find_key()	Combines shifts to recover the key.
vigenere_decrypt()	Decrypts ciphertext using the key.
vigenere_encrypt()	Encrypts plaintext for verification.
verify()	Checks whether re-encryption matches the original.

11. Results

Estimated Key Length: 14

Recovered Key: AMBROIETHOMAS

Verification: SUCCESS

12. Recovered Plaintext

DO YOU KNOW THE LAND WHERE THE ORANGE TREE BLOSSOMS
THE COUNTRY OF GOLDEN FRUITS AND MARVELOUS ROSES
WHERE THE BREEZE IS SOFTER AND BIRDS LIGHTER
WHERE BEES GATHER POLLEN IN EVERY SEASON
AND WHERE SHINES AND SMILES LIKE A GIFT FROM GOD
AN ETERNAL SPRINGTIME UNDER AN EVER BLUE SKY
ALAS BUT I CANNOT FOLLOW YOU TO THAT HAPPY SHORE
FROM WHICH FATE HAS EXILED ME
THERE IT IS THERE THAT I SHOULD LIKE TO LIVE
TO LOVE TO LOVE AND TO DIE

IT IS THERE THAT I SHOULD LIKE TO LIVE

IT IS THERE YES THERE

13. Conclusion

The project successfully demonstrates Vigenere Cipher cryptanalysis using Kasiski Examination, Index of Coincidence, frequency analysis and chi-square testing. The 14-character key AMBROISETHOMAS was recovered and the plaintext was correctly decrypted. Re-encryption produced the original ciphertext, giving a SUCCESS verification result.

14. How to Run

Compile:

```
g++ vigenere_attack.cpp -o vigenere_attack
```

Run:

```
./vigenere_attack
```